

LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

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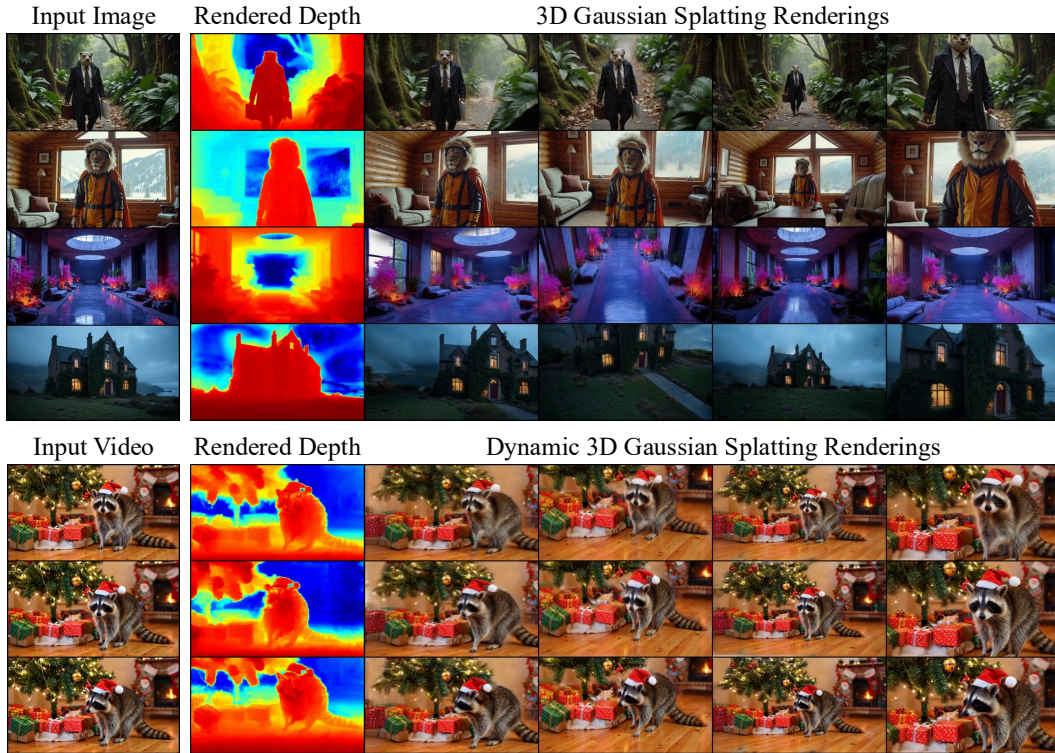


Figure 1: **Feed-Forward 3D and 4D Scene Generation.** From a single image (top), *Lyra* infers a 3D Gaussian Splatting (3DGS) representation in a feed-forward fashion, through self-distilling a video diffusion model without requiring real-world multi-view data. With a video input (bottom), *Lyra* infers a dynamic 3DGS that offers interactive control in both time (rows) and viewpoint (columns).

ABSTRACT

Explicit virtual environments are increasingly important for games, immersive media, and physical AI systems in robotics, autonomous driving, and industrial automation, where agents must render, navigate, and interact with coherent 3D worlds. Learning-based 3D reconstruction has made substantial progress, yet it typically relies on captured multi-view real-world data whose acquisition is costly, constrained, and difficult to scale across diverse scenes. Recent video diffusion models provide a complementary source of visual and geometric prior knowledge, but their outputs remain 2D videos rather than explicit representations suitable for real-time rendering and downstream simulation. We introduce *Lyra*, a self-distillation framework that transfers the camera-conditioned generative prior of a video diffusion model into an explicit 3D Gaussian Splatting (3DGS) representation without requiring real-world multi-view training data. *Lyra* augments the frozen video model with a 3DGS decoder that operates directly in the model’s latent space and learns from the synthesized outputs of the standard RGB decoder. This design converts synthetic, camera-controlled video supervision into feed-forward 3DGS prediction, enabling explicit scene generation from a text prompt or a single image

at inference time with real-time renderability. The same self-distillation principle extends to monocular video inputs, yielding time-conditioned dynamic 3DGS representations for 4D scene generation. Across the reported static and dynamic evaluation protocols, *Lyra* achieves state-of-the-art performance while preserving the practical advantage of an explicit, feed-forward scene representation.

1 INTRODUCTION

Scalable creation of high-quality 3D environments remains a fundamental problem in computer vision and graphics, with direct consequences for film-making, VR/AR, and physical AI systems that require closed-loop simulation. These settings do not merely require visually plausible images; they require explicit 3D representations that support real-time rendering, physical interaction, and consistent synthesis across viewpoints.

Neural 3D reconstruction methods (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) have substantially improved the recovery of such representations from posed images, but their dependence on accurate camera poses and high-quality captures limits their scalability. The limitation is amplified for dynamic scenes, where reconstruction commonly depends on synchronized multi-camera acquisition (Li et al., 2022). Moreover, reconstruction is intrinsically tied to observed content and therefore cannot synthesize plausible scene regions beyond the input views. Feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c) reduce some of this per-scene burden, yet they inherit a different bottleneck: diverse, large-scale 3D supervision is scarce, which weakens out-of-domain generalization.

Video diffusion models (Cosmos, 2025; Wan, 2025) provide a complementary source of supervision. Because they are trained on massive video corpora spanning varied environments and camera motions, including handheld trajectories, cinematic pans, and drone footage, these models encode substantial information about 3D scene structure without requiring explicit multi-view datasets. In contrast to reconstruction-only approaches (Charatan et al., 2024), generative video models can also hallucinate plausible content in unobserved regions. Their principal shortcoming is representational: they produce 2D frames rather than explicit 3D assets, which restricts their use in simulation and other downstream settings that demand geometric consistency, long-horizon coherence, and physical interaction. Recent camera-controlled video models (Wang et al., 2024e; Ren et al., 2025) show that video generators can be adapted to produce posed image sequences from comparatively small posed datasets, turning them into a strong source of both inputs and supervision for 3D reconstruction.

This paper unifies reconstruction and generation through *generative 3D scene reconstruction*. We introduce *Lyra*, a feed-forward method that converts the latent representations of a video diffusion model into explicit 3D environments in a single pass. Crucially, *Lyra* is trained by self-distillation (Fig. 2): a 3D Gaussian Splatting (3DGS) decoder operates directly in the video model’s latent space and learns from the model’s standard RGB decoder. Given a single image or video, we sample a camera trajectory, denoise the corresponding video latent, and decode it through two coupled branches. The RGB branch synthesizes the teacher video, while the 3DGS branch produces the student 3D representation. To expand viewpoint coverage, we sample multiple trajectories, generate multiple video latents, and train the 3DGS decoder to fuse the resulting information while reducing long-term and multi-view inconsistencies.

The resulting framework contributes three practical advantages that follow directly from this design. **(i)** It produces large-scale synthetic environments across diverse scenarios from video diffusion models, avoiding the need for real-world multi-view capture. **(ii)** By operating in latent space, it aggregates many views without the prohibitive memory cost of pixel-space feed-forward reconstruction. **(iii)** Its explicit 3DGS output provides geometrically coherent representations suitable for real-time rendering and downstream simulation. Consequently, at inference time, *Lyra* generates high-quality 3D Gaussian scenes from monocular inputs *without requiring any additional optimization or post-processing*.

We further extend the same self-distillation principle to dynamic 4D generation from monocular video. In this setting, the video model supplies space–time supervision, and the student learns time-conditioned 3DGS representations that support novel-view synthesis of dynamic scenes.

Overall, our work makes the following contributions:

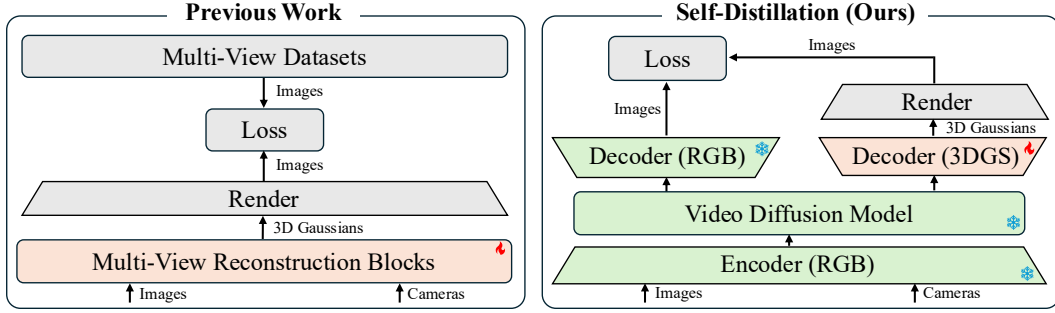


Figure 2: **Self-distillation framework of *Lyra***. Previous work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks using real-world datasets with limited diversity (Zhou et al., 2018; Ling et al., 2024b). In contrast, we propose a self-distillation framework (right) for generative 3D scene reconstruction. Precisely, a pre-trained camera-controlled video diffusion model with its RGB decoder output (teacher) supervises the rendering of the 3DGS decoder (student). Using a video model pre-trained on diverse 2D video data allows us to provide diverse multi-view supervision.

- We introduce a self-distillation framework that trains a 3DGS student decoder from a pre-trained camera-controlled video diffusion teacher, eliminating the need for captured multi-view real-world training data.
- We formulate latent-space feed-forward 3DGS reconstruction from multi-trajectory video latents, enabling efficient fusion of hundreds of synthesized views into a single explicit scene representation.
- We extend the framework to dynamic scenes, demonstrating 4D reconstruction from monocular video through time-conditioned 3DGS decoding.
- We show that the model generalizes across diverse scenes and achieves state-of-the-art results for single-image 3D scene generation and single-video 4D scene generation under the reported benchmarks.

2 RELATED WORK

Multi-view image generation. Early multi-view image generation methods (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) primarily addressed object-centric synthesis and generally omitted full scene backgrounds. Subsequent work (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) extended multi-view generation toward scene-scale content. CAT3D (Gao et al., 2024b), for example, adapts a pre-trained image diffusion model to generate multiple views from either generated or real images, after which the views are reconstructed into explicit representations such as NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023). Related space-time diffusion models further extend this strategy to dynamic 3D scenes observed from multiple viewpoints (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). These pipelines demonstrate the power of generative view synthesis, but explicit 3D grounding still depends on a costly, per-scene optimization stage that is not amortized. Our goal is therefore different: we directly generate an explicit 3D scene from text or a single image rather than first producing multi-view images and then reconstructing them by optimization.

Camera-conditioned video generation. Camera-conditioned video diffusion has made substantial progress in controllable scene synthesis. MotionCtrl (Wang et al., 2024e) introduced camera control by conditioning pre-trained video models on camera poses, and later methods (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) use Plücker coordinates to provide pixel-wise camera conditioning. A complementary line of work (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b) studies training-free camera control, while SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) show that large-scale synthetic data can produce strong camera-controlled video synthesis. Despite these advances, the generated outputs remain videos rather than explicit 3D assets. Our framework is orthogonal to these video-generation advances: it uses a camera-controlled video model as the teacher for feed-forward 3D generation, so the resulting scene can be rendered from novel viewpoints *after* generation. Concretely, we build on GEN3C (Ren et al., 2025) and distill its camera-controlled video latents into a 3DGS student decoder.

Feed-forward 3D models. Feed-forward 3D reconstruction initially focused on object-centric content and directly predicted NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) representations from text or images. More recent scene-level

methods (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) regress per-pixel 3D Gaussians from one or more images, but their training data are typically concentrated on datasets such as RealEstate10K (Zhou et al., 2018), which limits transfer to generated and out-of-distribution scenes. Recent generative reconstruction methods address this gap. Bolt3D (Szymanowicz et al., 2025b) trains a pointmap (Wang et al., 2024b) autoencoder to generate pointmaps jointly with multi-view images for feed-forward reconstruction, and Wonderland (Liang et al., 2025a) uses a camera-controlled video model with a feed-forward network to produce 3D Gaussians. In contrast, *Lyra* explicitly formulates the video model as a self-distillation teacher and trains the student 3D model *without requiring real-world multi-view data*. The same formulation also supports feed-forward 4D scene generation, a setting that remains largely unexplored.

3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

Lyra converts the implicit 3D knowledge of camera-controlled video diffusion models into an explicit 3DGS decoder capable of producing high-quality scene representations. The method is organized as a teacher–student framework: the video diffusion model serves as the teacher, its RGB videos provide supervision, and the student 3DGS decoder operates in the same latent space, as illustrated in Fig. 2. This section first specifies the video diffusion backbone (Sec. 3.1) and then describes its use in self-distillation for the 3DGS decoder (Sec. 3.2). We subsequently motivate the decoder design for latent-space feed-forward reconstruction (Sec. 4) and describe the minimal modifications that extend the pipeline to dynamic 3D scenes (Sec. 5).

3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Because the 3DGS decoder receives supervision only from the camera-controlled video diffusion model, the teacher must provide geometrically consistent videos under prescribed camera motion. We therefore build on GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model designed for 3D-consistent generation. The following overview focuses on the GEN3C components that are essential for the self-distillation pipeline.

Spatiotemporal 3D cache. GEN3C improves video consistency and camera-control precision by constructing a spatiotemporal 3D cache $\{\mathbf{P}^{t,v}\}$ from input image(s) or videos. Each $\mathbf{P}^{t,v}$ is a colored point cloud obtained by unprojecting the depth estimation (Wang et al., 2024a) of an RGB image captured from camera viewpoint v at time t . The cache is organized as an $L \times V$ array, where L denotes the temporal length and V denotes the number of camera views.

Rendering and structured guidance. GEN3C uses the cache by rendering each point cloud from arbitrary specified camera poses, producing RGB images $\mathbf{I}^{t,v}$ and disocclusion masks $\mathbf{M}^{t,v}$ via $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$, where $\mathcal{R}$ denotes the rendering function that projects the 3D point cloud $\mathbf{P}^{t,v}$ onto the 2D camera plane according to the camera pose $\mathbf{C}^t$ at time t . Given a sequence of camera poses $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$, rendering all cache elements produces V videos of length L , which are stacked into image sequences $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$ and mask sequences $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$ for each view v . The disocclusion masks identify regions that the video diffusion model must synthesize, and the rendered RGB/mask sequences provide structured visual guidance for subsequent video generation.

Video variational autoencoder (VAE). Diffusion-based video generation models commonly use a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) to compress videos into a lower-dimensional latent space, thereby reducing the cost of training and inference. Given a RGB video $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$, the pre-trained VAE encoder $\mathcal{E}$ maps the video to $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$, and the diffusion process is performed in this latent space. The final video $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$ is decoded with the pre-trained VAE decoder $\mathcal{D}_{rgb}$. We adopt the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model, for which the latent channel dimension is $C = 16$, the temporal dimension is $L' = (L - 1)/\tau + 1$, and the spatial dimensions are $h = H/\sigma$ and $w = W/\sigma$, with temporal compression factor $\tau = 8$ and spatial compression factor $\sigma = 8$.

3.2 SELF-DISTILLATION

We train the 3DGS decoder $\mathcal{D}_s$ as a student in a teacher–student paradigm, with the camera-controlled video diffusion model $\mathcal{V}$ serving as the teacher. To obtain diverse synthetic supervision, we curate a

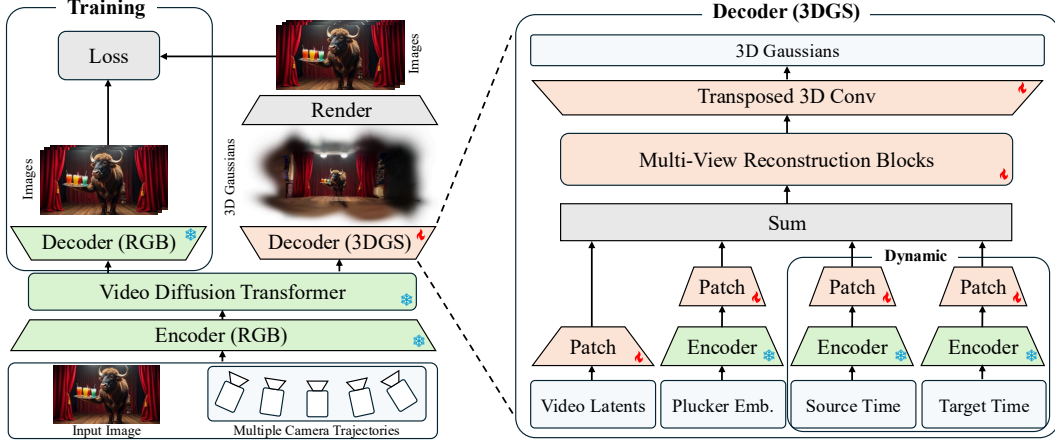


Figure 4: **3D Generative Reconstruction Framework.** Our pipeline builds upon a camera-controlled video diffusion model (Ren et al., 2025) pre-trained on large scale data. We train a 3D Gaussian Splatting (3DGS) decoder by aligning the 2D image renderings of generated 3DGS scenes with the RGB-decoded generations of the pre-trained video model. We only train the 3DGS decoder while freezing the pre-trained autoencoder and diffusion model. At inference time we directly use the 3DGS decoder, without requiring the RGB decoder anymore. Time conditioning within the 3DGS decoder allows us to easily extend our approach from static to dynamic 3D scene generation.

large-scale prompt set with large language models (OpenAI, 2025; Bai et al., 2023), generate images I with an image diffusion model (Black Forest Labs, 2024), and expand those images into multi-view sequences using GEN3C (Ren et al., 2025).

Teacher-student setup. Given an input image I and a sampled camera trajectory $\{\mathbf{C}^t\}_{t=1}^L$, the video diffusion model $\mathcal{V}$ generates a denoised video latent $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$. The latent $\mathbf{z}$ is decoded along two branches: the pre-trained RGB decoder $\mathcal{D}_{rgb}$ produces video frames $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$, while the 3DGS decoder $\mathcal{D}_s$ outputs explicit 3D Gaussians $\mathbf{G}$. Rendered views from 3DGS (*student*) are defined as $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$ and are supervised to match RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}$ (*teacher*), forming the self-distillation loop.

Multi-trajectory supervision. In our paper, to enlarge viewpoint coverage for the input image, we sample $V = 6$ camera trajectories per input image, as shown in Fig. 3. For each trajectory v , we construct a spatiotemporal cache $\{\mathbf{P}^{t,v}\}_{t=1}^L$ along with $L = 121$ camera poses $\{\mathbf{C}^{t,v}\}_{t=1}^L$. Passing these caches through video diffusion model $\mathcal{V}$ yields latents $\mathbf{z}^v$, which the teacher decodes into RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$. The 3DGS decoder $\mathcal{D}_s$ learns to fuse these multiple $\mathbf{z}^v$ into coherent Gaussians $\mathbf{G}$, while filling disoccluded regions. Through this self-distillation design, the 3DGS decoder $\mathcal{D}_s$ is trained entirely from synthetic supervision provided by $\mathcal{V}$. Operating in latent space enables efficient aggregation of multiple trajectories, while the explicit Gaussian output $\mathbf{G}$ enforces geometric consistency and supports downstream simulation and real-time rendering.

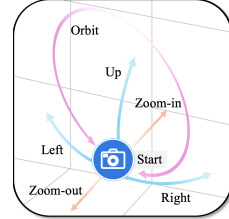


Figure 3: Sampled six camera trajectories to maximize view coverage.

4 FEED-FORWARD RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

Our feed-forward 3DGS decoder $\mathcal{D}_s$ is designed to transform the synthesized multi-view latents generated by our video model $\mathcal{V}$ into an explicit 3D representation that can be rendered from arbitrary viewpoints. We choose 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) due to its explicit representation and fast rendering speed. We visualize our detailed 3DGS decoder in Fig. 4 (right).

4.1 3DGS DECODER

Scaling Latent-Based 3D Reconstruction. Previous feed-forward reconstruction frameworks generate 3D Gaussians for each pixel, and thus, are limited by the number and resolution of input views they can handle. For example, GS-LRM (Zhang et al., 2024e) operates on 2–4 images at

512 $\times$ 512 resolution, while AnySplat (Jiang et al., 2025a) is trained with 24 images at 448 $\times$ 448 resolution. Our video model synthesizes $V \times L = 6 \times 121 = 726$ input views at 704 $\times$ 1280 resolution—far beyond the capacity of existing methods, which exceeds GPU memory limit. The bottleneck lies in the attention mechanism applied to visual tokens, whose memory and compute requirements grow with the number of pixels. To overcome this limitation, we avoid scaling in pixel space and instead operate directly in the compressed video latent space produced by our camera-controlled video diffusion model. The multi-view video latents are denoted as $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$.

Architecture. The 3DGS decoder $\mathcal{D}_s$ first maps its inputs into the hidden dimension of the main reconstruction blocks and outputs per-pixel 3D Gaussian features $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$. Specifically, the inputs are the multi-view video latents $\mathbf{Z}$ and encoded Plücker embeddings $\mathbf{E}$, both with latent dimensions $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. Each input component is first patchified to match the hidden dimension of the main reconstruction blocks. We do not require any additional visual encoder—only a spatial 2 $\times$ 2 patchification layer (Dosovitskiy et al., 2021) to transform the video latents into flattened tokens for the reconstruction network. The sum of both inputs is processed through the reconstruction blocks. We follow the design introduced in Long-LRM (Ziwen et al., 2024), i.e., one block consists of one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers. We repeat the block twice to get 16 layers with 512 hidden dimensions. Finally, a transposed 3D convolution maps the hidden representation to 14 Gaussian channels: 3D position (x, y, z) , scale (s_x, s_y, s_z) , rotation quaternion (q_w, q_x, q_y, q_z) , opacity α , and RGB (r, g, b) . Formally, the static decoder is expressed as $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$.

Plücker embeddings. Raw Plücker embeddings are first computed from camera poses $\{\mathbf{C}^{t,v}\}$ and intrinsics as $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$. We reuse the RGB encoder $\mathcal{E}$ from the pre-trained VAE to encode Plücker embeddings. Specifically, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins using $\mathcal{E}$, and concatenate latent along the channel dimension to obtain $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$. A lightweight MLP maps $\mathbf{E}^{\text{enc}}$ to $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, reducing the channel dimension by a factor of 2 to match video latent $\mathbf{Z}$.

4.2 LOSS FUNCTION

Image-based supervision. We supervise our 3DGS decoder with an image-based reconstruction loss. The reconstruction loss is split into a Mean Squared Error (MSE) loss $\mathcal{L}_{mse}$ and an LPIPS loss $\mathcal{L}_{lpips}$ (Zhang et al., 2018) using VGG (Simonyan & Zisserman, 2014) as a feature extractor.

Depth supervision. We observed that the decoder trained with only RGB loss often produces flattened geometry. To address this, we additionally supervise the rendered depth maps using the consistent video depth estimated by an off-the-shelf system ViPE (Huang et al., 2025), and the scale-invariant depth loss $\mathcal{L}_{depth}$ from Long-LRM (Ziwen et al., 2024).

Opacity-based pruning. Similar to Long-LRM (Ziwen et al., 2024), we use an L1 regularization on the opacity $\mathcal{L}_{opacity}$ and remove the Gaussians with the lowest 80% opacity.

Total loss. Our total loss is computed as the weighted sum of all losses with weight factors λ_i

$$\mathcal{L} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{lpips} \mathcal{L}_{lpips} + \lambda_{depth} \mathcal{L}_{depth} + \lambda_{opacity} \mathcal{L}_{opacity} \quad (1)$$

where we set $\lambda_{mse} = 1.0$, $\lambda_{lpips} = 0.5$, $\lambda_{depth} = 0.05$, and $\lambda_{opacity} = 0.1$.

5 EXTENSION TO DYNAMIC 3D SCENES

Our approach can be extended to handle dynamic 3D scenes with minimal changes, outlined below.

Self-distillation for dynamic scenes. Our dynamic 3D setup closely follows the design of the static 3D counterpart. Instead of a single image, the video model takes a single video of length L with corresponding camera poses as input and generates multi-view video latents capturing the same underlying motion. For video inputs, we follow a protocol similar to the static case: sampling diverse text prompts with large language models, then generating videos using video diffusion models (Cosmos, 2025; Wan, 2025). The generated videos are then annotated with camera poses and depth maps using ViPE (Huang et al., 2025).

Dynamic 3DGS decoder architecture. We follow the bullet-time design of Liang et al. (2025b) and design our dynamic 3DGS decoder $\mathcal{D}_s$ to generate time-dependent 3D Gaussians for all the video

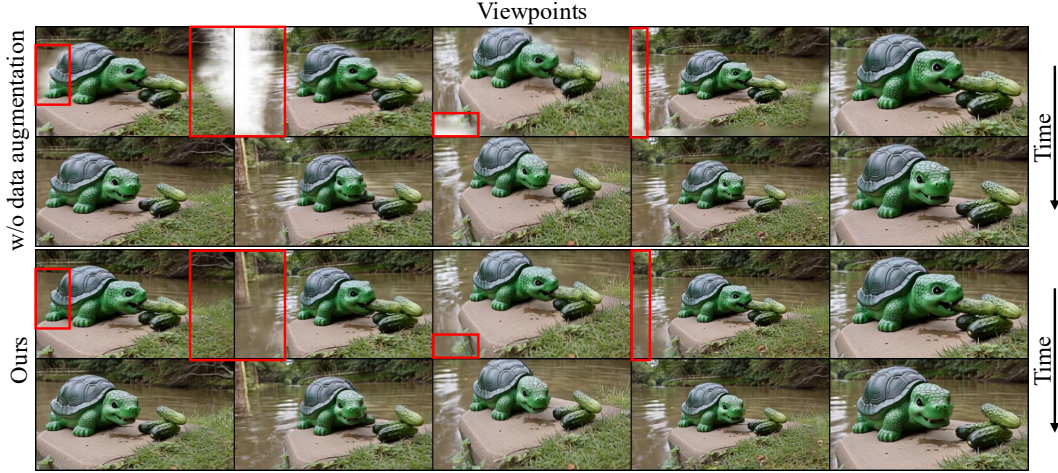


Figure 5: **Dynamic data augmentation.** When naively supervising the time-aware 3DGS decoder, we observe artifacts in the generated 3D Gaussians. Specifically, early timesteps from extreme poses exhibit low opacity in regions not covered by the supervision signal. To address this, we augment the supervision data with a motion-reversed video, ensuring that each timestep is observed from the full spatial coverage, thereby preventing low opacity artifacts in the early timesteps.

frames. Specifically, the dynamic 3DGS decoder $\mathcal{D}_s$ closely follows the architecture of $\mathcal{D}_s$, except that we augment the $\mathcal{D}_s$ input with the encoded source and target time embeddings $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The unencoded (raw) source times are assigned to each input frame, forming $\mathbf{T}^{\text{src, raw}} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$, and the target time is $\mathbf{T}^{\text{tgt, raw}} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$. Both are spatially repeated and augmented with a 2-dimensional sinusoidal embedding to expand the channel dimension to 3, then encoded with the RGB encoder $\mathcal{E}$ to latent dimensions $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The target time is repeated along the latent temporal dimension to match L' . During training, a target time step is randomly sampled, and the corresponding 3D Gaussians are decoded and supervised across all trajectories generated with dynamic data augmentation. We fine-tune $\mathcal{D}_d$ from a pre-trained $\mathcal{D}_s$ with the patchification layers of $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ initialized to zeros. $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ are directly added to the sum of $\mathbf{Z}$ and $\mathbf{E}$. Formally, the dynamic decoder is defined as $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}})$.

Dynamic data augmentation. In the static 3D scenes, where time is not a factor, frames from different timesteps can all be used to supervise the 3DGS renderings. In contrast, for dynamic 3D scenes, the 3DGS changes over time, so only frames from the corresponding timestamp are valid for supervision. This restriction can lead to a trivial solution, where the 3DGS at a given timestep simply ignores information from frames at other timesteps (see Fig. 5).

To mitigate this issue, we introduce a dynamic data augmentation strategy that balances supervision across timesteps. Early frames are naturally associated with viewpoints close to the input image, whereas later frames correspond to more distant viewpoints. To counter this imbalance, we augment the training data with paired supervision views for each timestep — one near and one far. Concretely, we reverse the input video in frame order and feed it into the video model $\mathcal{V}$, which produces six additional multi-view sequences. After reversing these sequences back to the original motion order, we obtain six trajectories that progress inward (from far viewpoints toward the input). Combined with the original six outward trajectories, this yields 12 supervision views per timestep. Importantly, this augmentation is applied only during training: the reversed trajectories act purely as extra supervision signals to prevent collapse from pruning artifacts and are not required at inference time.

6 EXPERIMENTS

6.1 EXPERIMENTAL SETUP

Datasets. Training uses no existing multi-view datasets. Instead, we construct the *Lyra* dataset, in which supervision for the 3DGS decoder is supplied solely by the video model. The static 3D reconstruction setting contains 59,031 images, and the dynamic 4D setting contains 7,378 videos. All samples originate from diverse text prompts spanning indoor and outdoor environments, humans,

Table 1: **State-of-the-art comparisons.** We compare our method with previous works for single image to 3D generation using RealEstate10K, DL3DV, and Tanks-and-Temples datasets.

Method	RealEstate10K			DL3DV			Tanks-and-Temples		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ZeroNVS	13.01	0.378	0.448	13.35	0.339	0.465	12.94	0.325	0.470
ViewCrafter	16.84	0.514	0.341	15.53	0.525	0.352	14.93	0.483	0.384
Wonderland	17.15	0.550	0.292	16.64	0.574	0.325	15.90	0.510	0.344
Bolt3D	21.54	0.747	0.234	-	-	-	-	-	-
Ours	21.79	0.752	0.219	20.09	0.583	0.313	19.24	0.570	0.336



Figure 6: **Image-to-3DGS Generation.** We visualize five views from generated 3DGS scenes.

animals, and both realistic and imaginative content. For each image in the 3D setting, and each video in the 4D setting, we synthesize six camera trajectories, producing 354,186 videos for 3D training and 44,268 videos for 4D training.

Baselines. We compare against state-of-the-art methods for generative single-image-to-3D reconstruction: ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b). Because *no source code is available* for evaluating these methods on our out-of-distribution evaluation set, the principal quantitative comparisons use the values reported in the corresponding papers.

Evaluation. Following prior work, we evaluate single-image-to-3D generation on RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks and Temples (Knapitsch et al., 2017). The protocol follows Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b), and performance is measured with standard reconstruction metrics: PSNR, SSIM, and LPIPS.

6.2 MAIN RESULTS

Quantitative results. Tab. 1 reports single-image-to-3D quantitative results across the three evaluation benchmarks. *Lyra* ranks first on every reported dataset and metric: it achieves the best PSNR, SSIM, and LPIPS on RealEstate10K, DL3DV, and Tanks-and-Temples under the shared evaluation protocol. These results show that self-distillation from camera-controlled video generation provides effective supervision for explicit 3D scene reconstruction. Because the reconstructed scenes are learned from the teacher video model, further improvements in camera-controlled video generation are expected to directly strengthen the reconstruction quality. Appendix C provides additional quantitative comparisons on the *Lyra* dataset.

Qualitative results. Fig. 1 and Fig. 6 visualize novel-view renderings from generated 3DGS scenes; video results will be released upon acceptance. Across static and dynamic examples, *Lyra* synthesizes plausible unseen regions while preserving consistency with the input image or video. Appendix C includes qualitative comparisons on the *Lyra* dataset.



Figure 7: **Ablations.** We visualize ablation results by rendering the same extreme novel viewpoint after image-to-3DGS generation; depth visualizations and ablations are provided in Appendix C.4.

6.3 ABLATIONS

Tab. 2 isolates the design choices that enable the full model on diverse out-of-distribution prompts. In these ablations, 3DGS renderings are compared with videos generated by the camera-controlled video diffusion model, so each row measures how a component affects the fidelity of the student representation relative to its teacher supervision.

Real data only. Replacing self-distillation with real multi-view datasets, namely RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b), substantially reduces performance on diverse prompts. This outcome reflects the limited out-of-distribution coverage of commonly used real datasets relative to the synthetic supervision generated by the video teacher.

Self-distillation + real data. Joint training with self-distillation and real data does not improve over self-distillation alone. The result indicates that, for this setting, the synthetic teacher supervision is sufficiently diverse and consistent to train the reconstruction model without additional real multi-view data.

No depth loss. Removing the depth loss weakens geometric structure and slightly reduces image-based metrics. Depth visualizations in Appendix C.4 further illustrate its role in preventing flattened reconstructions.

No opacity pruning. Without opacity pruning, all Gaussians are retained. Learning to prune the lowest-opacity Gaussians produces a more compact representation and slightly improves visual quality. At $H = 704$ and $W = 1280$, rendering takes 18ms with pruning and 30ms without pruning, corresponding to a $1.67\times$ speedup.

No LPIPS loss. The LPIPS loss improves robustness to input inconsistencies and helps preserve high-frequency details, yielding stronger perceptual reconstruction quality.

No multi-view fusion. A simpler alternative would generate 3D Gaussians independently for each camera trajectory and then merge the resulting Gaussians into one point cloud. Learned multi-view fusion through the reconstruction blocks performs substantially better because tokens from different trajectories attend to one another before the final 3DGS prediction.

No Mamba-2. Replacing the joint Transformer/Mamba-2 reconstruction blocks with Transformer-only blocks slightly reduces quality and substantially increases runtime. A forward pass with $V = 6$, $L = 121$, $H = 704$, and $W = 1280$ takes 3213ms with the joint blocks, compared with 20922ms for Transformer-only blocks, giving a $6.5\times$ speedup.

No latent-based 3DGS. Pixel-space feed-forward methods such as BTimer (Liang et al., 2025b) are restricted to 12 input frames, whereas the latent-space decoder can process up to 726 input frames. Moving the 3DGS decoder from latent space to pixel space therefore exceeds memory limits in this setting.

Table 2: **Ablation study on *Lyra* dataset.**

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	24.77	0.837	0.224
Data			
real data only	19.08	0.659	0.413
self-distill. + real data	24.74	0.823	0.236
Loss			
w/o depth loss	24.31	0.811	0.247
w/o opacity pruning	24.55	0.820	0.237
w/o LPIPS loss	23.74	0.766	0.370
Architecture			
w/o multi-view fusion	17.73	0.632	0.446
w/o Mamba-2	24.58	0.818	0.241
w/o latent 3DGS		OOM	

7 CONCLUSION

We presented *Lyra*, a unified framework for static 3D and dynamic 4D generation from monocular inputs. Rather than collecting multi-view datasets, the method uses a camera-controlled video diffusion model as a teacher and trains a student 3DGS decoder by self-distillation. The decoder operates in the latent space of the video model and directly predicts 3D Gaussians, avoiding per-scene optimization and post-processing while retaining the explicit representation needed for real-time rendering. The static and dynamic results demonstrate that video diffusion models can serve as effective supervision engines for feed-forward 3D and 4D scene generation across diverse content.

The scale and consistency of the generated scenes are currently bounded by the capacity of the camera-controlled video diffusion teacher. Improvements in such video models should therefore translate directly into larger and more consistent scene synthesis. Future work may also adapt auto-regressive techniques (Chen et al., 2024a) to this framework for large-scale generation, and may incorporate motion and tracking information within the reconstruction network, as in concurrent work (Lin et al., 2025), to further improve dynamic visual quality.

8 ETHICS STATEMENT

As with generative AI systems more broadly, *Lyra* could be misused to create misleading or synthetic 3D/4D content. The objective of this research, however, is to advance simulation, robotics, and embodied AI by providing scalable and controllable tools for generating explicit environments. We expect the positive impact of the work to arise primarily through the following capabilities and insights.

- **The ability to enable realistic simulation for embodied agents:** The method provides controllable and physically consistent camera motion within generated 3D and 4D environments, supporting the training and evaluation of agents that must perceive, navigate, and interact with complex scenes.
- **Improved scalability for data engines:** By removing the need for multi-view capture and expensive per-scene optimization, the method supports high-fidelity, customizable scene generation at scale, which is particularly relevant for robotics, reinforcement learning, and closed-loop simulation research.
- **Better technical understanding:** As an academic study, the work clarifies how camera control, spatiotemporal generative models, and explicit 3D reconstruction can be combined, offering useful evidence for future research in computer vision, graphics, and embodied AI.

We therefore caution against deceptive or wrongful uses of realistic generative outputs, particularly when the resulting media could be misleading or unverifiable. Safeguards such as provenance tracking, dataset documentation, and careful evaluation practices remain important for ensuring that generative models advance science and society responsibly.

9 REPRODUCIBILITY STATEMENT

To enable full reproduction of the reported results and to support future work on generative 3D and 4D reconstruction, we will release the training and inference code, model weights, and data for both 3D and 4D generation upon acceptance.

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A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

This section provides additional information on the GEN3C (Ren et al., 2025) video diffusion model backbone introduced in Sec. 3.1 and how we further improve its 3D consistency.

Conservative mask refinement. A key limitation of GEN3C’s forward warping is background leakage, where occluded object parts in the source view (e.g., lion body in Fig. 8) become visible in novel viewpoints but are not properly indicated in the disocclusion mask. This leads to incomplete foreground completion by the video diffusion model, as shown in Fig. 8 (second and fourth columns).

To address this, we adopt a conservative strategy: since the geometry of occluded regions in $\mathbf{P}^{t,v}$ is unknown, we assume all such areas are occupied and derive refined disocclusion masks accordingly. This allows the video model to reason about potentially visible foreground regions that were missed by standard point-based rendering.

Our approach constructs a triangular mesh from camera-space points $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$ by connecting spatially adjacent pixels, similar to Hu et al. (2021):

$$\mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

where $\mathbf{p}_{u,v}$ denotes the 3D point at pixel (u, v) in $\mathbf{P}_c^{t,v}$. This surface mesh serves as a boundary between visible and invisible regions. For each rendered pixel, we compare the standard point-based depth $\mathbf{D}^{t,v}$ with the mesh-interpolated depth $\mathbf{D}_{\mathcal{M}}^{t,v}$ obtained via ray-surface intersection:

$$\mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

where $\mathbf{M}_{\text{orig}}^{t,v}$ denotes the original disocclusion mask from forward warping and ϵ is a small tolerance factor. When the mesh surface is closer than the original points, we conservatively mask out these regions as potential foreground disocclusions. This produces more reliable guidance for video generation, especially under large viewpoint changes, as shown in Fig. 8 (third and fifth columns).

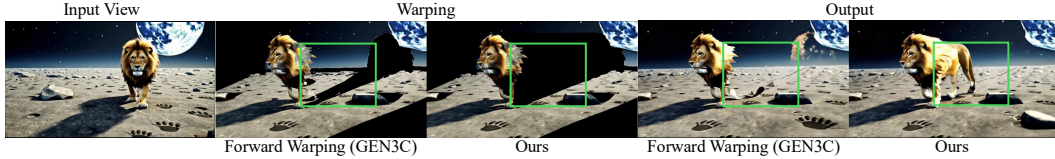


Figure 8: **Rendering function comparison.** Our model builds upon GEN3C (Ren et al., 2025) that uses forward warped images as camera control conditioning. We visualize forward warping (left) vs. our conservative rendering function (right). The highlighted region is incorrectly filled with background pixels in forward warping but properly masked in ours, enabling correct completion.

B MODEL DETAILS

Progressive training. Directly training the 3DGS decoder on 6 trajectories of 121 frames at 704×1280 is expensive, hence we train our network progressively, as shown in Tab. 3. Furthermore, we use the final static pre-trained model to initialize our dynamic model. The total training time takes 6 days with 8 NVIDIA A100 80 GB GPUs. We use gsplat (Ye et al., 2025) as the 3DGS implementation.

Subsampling Gaussians. Generating pixel-aligned 3D Gaussians scales linearly with each spatial and temporal resolution. Our generated scenes represent 726 frames at 704×1280 resolution, leading to 654,213,120 per-pixel Gaussians. Hence, we only generate one 3D Gaussian per 8×8 spatial neighborhood, reducing the total number of 3D Gaussians by a factor of 64, i.e., 10,222,080. Moreover, our opacity-based pruning further reduces the number to 2,044,416, making the output more compact.

Joint Transformer-Mamba blocks. We use the reconstruction block design introduced in Long-LRM (Ziwen et al., 2024), i.e., one block consists of one Transformer layer followed by seven

Table 3: **Progressive training setup.**

Stage	$H \times W$	L	V	S	B	Steps
Static						
1	176×320	17	1	17	4	10k
2	176×320	49	1	49	4	2.5k
3	352×640	49	1	49	2	2.5k
4	704×1280	49	1	49	1	2.5k
5	704×1280	121	1	9	1	57.5k
6	704×1280	121	1-6	9	1	7k
Dynamic						
7	704×1280	121	6	12	1	10k

Mamba-2 layers. We repeat the block twice to get 16 layers with 512 hidden dimensions. Note that our network is significantly smaller than the one used in Long-LRM (Ziwen et al., 2024) or Wonderland (Liang et al., 2025a), which use 24 layers with 1024 hidden dimensions. While pure Transformers are typically more powerful, they are significantly slower in training and inference than Mamba2. Since we trained both configurations within the same training budget, we hypothesize that transformer-only blocks would need considerably more training iterations to fully catch up in rendering quality.

Camera encoding. We use Plücker embeddings for camera encoding, following prior works (Kant et al., 2024; Zhang et al., 2024e). The embeddings are computed at full spatial and temporal resolution and then projected into the video latent space using the pre-trained RGB encoder. Since the RGB encoder expects 3 input channels but Plücker embeddings are 6-dimensional, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins. The resulting latent encodings are then concatenated along the channel dimension.

Time encoding. Similarly, to make the 1-dimensional time input compatible with the 3-channel input expected by the RGB encoder, we concatenate a 2-dimensional sinusoidal embedding to the original time value along the channel dimension. We then replicate these values across the spatial dimensions before encoding the time into the video latent space. This approach allows us to independently encode both the source and target times into the compressed latent representation.

Dynamic data augmentation. We visualize our dynamic data augmentation procedure for two example camera trajectories in Fig. 9. The procedure creates pairs of supervision views that cover the scene from the same motion state but different extreme viewpoints.

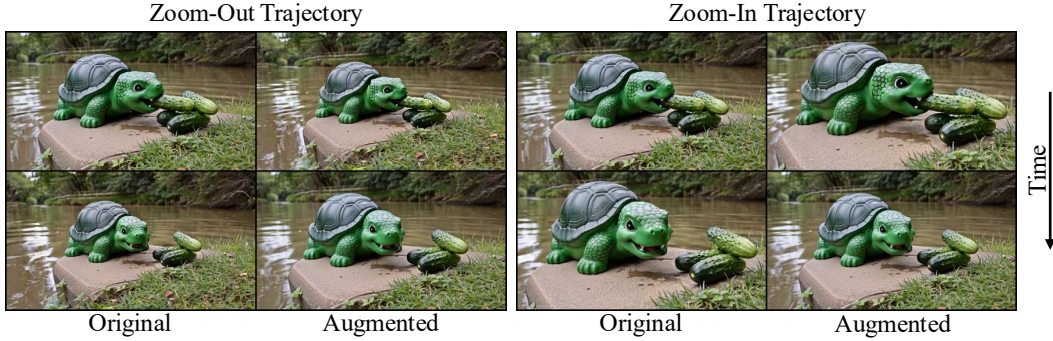


Figure 9: **Dynamic data augmentation videos.** We augment the supervision data with a motion-reversed video, ensuring that each timestep is observed from the full spatial coverage, thereby preventing low opacity artifacts in the early timesteps. We show two example trajectories, i.e., zoom-out and zoom-in, and visualize their corresponding augmented videos. The augmented videos are flipped in their camera motion.



Figure 10: **Image-to-3DGS Comparison.** We compare our method with BTimer (GEN3C) and visualize five views from generated scenes. We observe significantly fewer artifacts and higher fidelity.

C ADDITIONAL EVALUATION

C.1 ADDITIONAL BASELINE DETAILS

We additionally compare our model against BTimer (Liang et al., 2025b), a recent approach for joint 3D and 4D reconstruction from posed images or videos. We reached out to the authors of BTimer to conduct experiments for us using the same scenes as *Lyra*. Since BTimer is purely regression-based, we integrate it with our GEN3C (Ren et al., 2025) video diffusion backbone. Specifically, we use the same camera trajectories as in our model and provide RGB-decoded videos from GEN3C as input to BTimer. BTimer operates in pixel space and requires 12 input images. However, as discussed in the main paper, pixel-space models such as BTimer run out of memory when directly using the 726 high-resolution frames generated by GEN3C. To address this, we uniformly subsample GEN3C frames. For static evaluations, we select 12 frames that evenly cover the available viewpoints. For dynamic evaluations, we sample 11 frames to span the viewpoint range and additionally include the bullet-time frame corresponding to the motion being reconstructed. This ensures that the target motion state is always present as a reference. We denote this integrated baseline as BTimer (GEN3C), i.e., BTimer combined with GEN3C.

C.2 ADDITIONAL STATIC 3D EVALUATION

For the task of image to 3DGS generation, we show qualitative comparisons with BTimer (GEN3C) in Fig. 10 and qualitative results in Tab. 4a. We observe higher quality and less artifacts for our method.

C.3 DYNAMIC 3D EVALUATION

BTimer (GEN3C) comparison. For dynamic 3D evaluation, we compare our method with BTimer (GEN3C), as discussed in Sec. C.1, on 100 out-of-distribution videos from our dynamic *Lyra* dataset. We crop the video to 512×512 for fair comparisons, as BTimer was mainly trained on that resolution. We show results in Tab. 4b and observe that our method significantly outperforms BTimer (GEN3C).

Table 4: Quantitative results on static and dynamic *Lyra* datasets.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	16.32	0.580	0.427	BTimer (GEN3C)	20.29	0.687	0.315
Ours	24.92	0.834	0.183	Ours	23.07	0.779	0.231

(a) Comparison on static *Lyra* dataset. (b) Comparison on dynamic *Lyra* dataset.

C.4 DEPTH VISUALIZATION

We visualize depths from our generated 3DGS in Fig. 11. Using depth supervision prevents flat geometries without sacrificing visual quality.

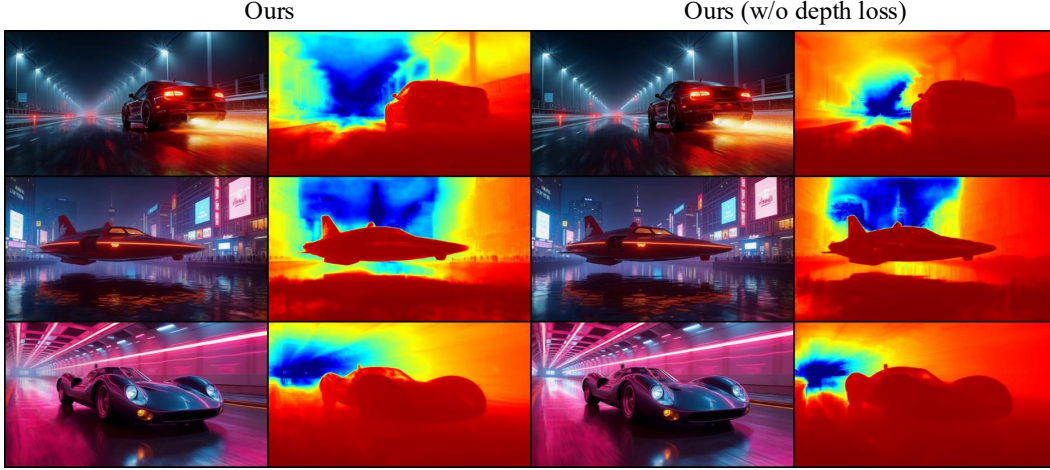


Figure 11: **Depth loss ablation.** We visualize 3DGS renderings and corresponding depth maps. Using the depth loss as additional supervision prevents flat geometries without sacrificing visual quality.

D ROBOT SIMULATION

One of our primary motivations for building a diverse 3D generator is its ability to synthesize simulation environments for autonomous agents. To this end, we develop an end-to-end pipeline: we first generate 3D scenes from text with our model, export the 3D Gaussians as a .ply file, and convert the .ply files into a .usdz format. We use 3DGUT (Wu et al., 2025a) and its export function to create .usdz files. The resulting .usdz file can then be imported into the NVIDIA Isaac robot development platform as a physically based virtual environment. AI-based robots can be trained and tested under diverse conditions within these environments. Early demo results will be released upon acceptance; we include a screenshot of the interface in Fig. 12.

E RELATED WORK

In this section, we will discuss additional related work. We highlight the differences to previous works CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b) in Fig. 13.

3D generation. Early approaches to 3D generation primarily focused on single object categories, extending GANs into the 3D domain by leveraging neural renderers as an inductive bias (DeVries et al., 2021; Chan et al., 2022; Or-El et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). With the introduction of CLIP-based supervision (Radford et al., 2021), the field advanced toward more flexible and diverse asset creation, enabling both text-driven generation and editing (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). More recently, diffusion models have significantly improved quality by replacing CLIP guidance with Score Distillation Sampling (SDS) (Poole et al., 2023; Wang et al., 2023b; Lin et al., 2023a; Chen et al.,

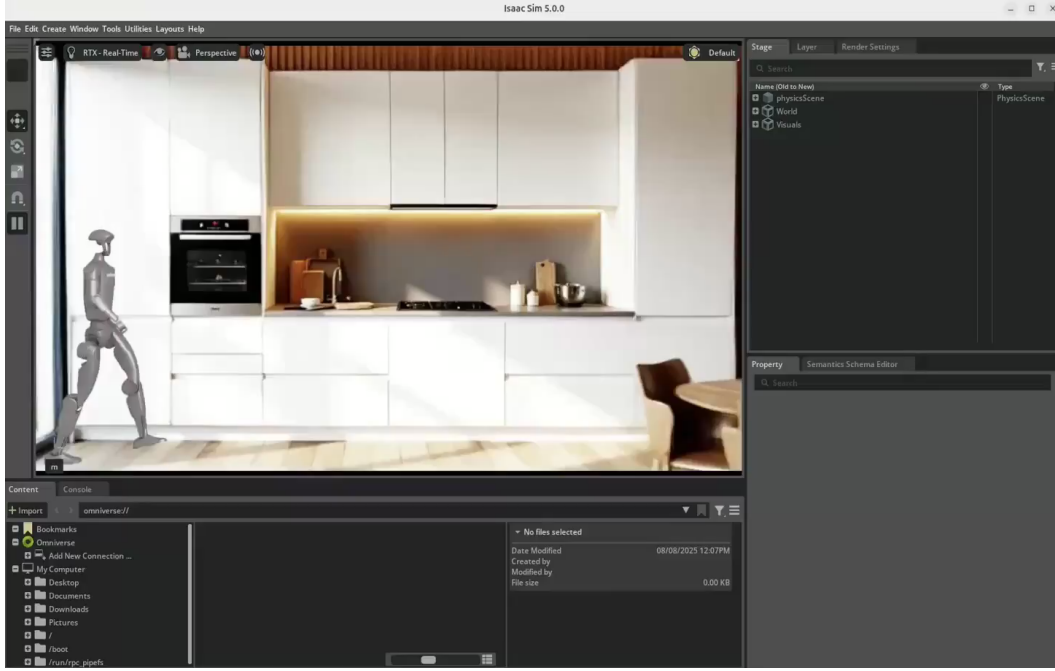


Figure 12: **Robot Simulation.** We visualize a frame of a robot simulation within the Isaac Sim 5.0 simulation framework that takes a generated 3DGS scene from our method as input.

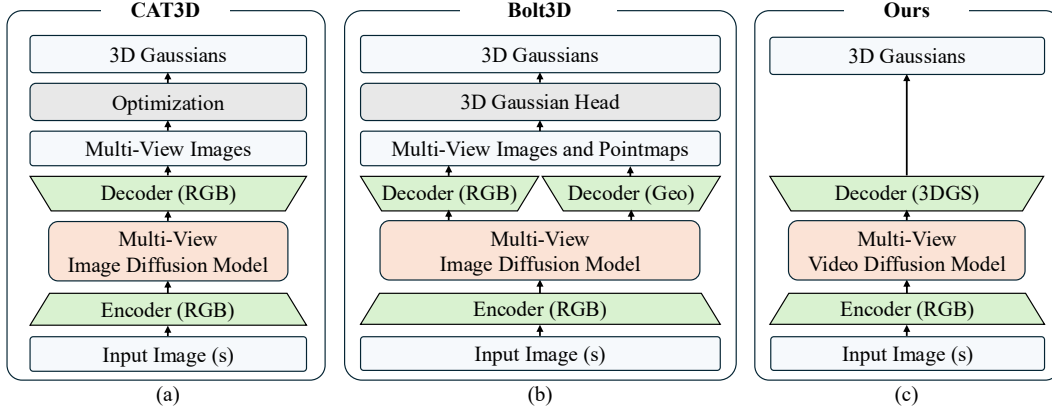


Figure 13: **Approaches for 3D Generation.** CAT3D (Gao et al., 2024b) proposed a multi-view image diffusion model that outputs images from novel viewpoints; subsequently, the images are reconstructed into 3D Gaussians using optimization. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to output pointmaps using a geometry autoencoder; instead of relying on optimization, the 3D scene is predicted with a feed-forward 3D Gaussian head. In contrast, our work builds upon a pre-trained camera-controlled video diffusion model and *directly* decodes the multi-view video latents into 3D Gaussians.

2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a).

To further enhance structural coherence, several methods enforce multi-view consistency by generating scenes from different perspectives (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a; Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024; Tang et al., 2024c; Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b), while others adopt iterative inpainting as a strategy for scene synthesis (Höllein et al., 2023; Shriram et al., 2024). Another line of work explores lifting 2D images into 3D using NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or mesh-based representations coupled with diffusion

models (Chan et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024).

Feed-forward 3D models. Complementary efforts investigate fast, feed-forward techniques that directly predict 3D content from images or text inputs (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e;d; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). However, in contrast to our method, these approaches remain limited to producing static 3D scenes. Other methods focus on specific scenes such as faces (Kirschstein et al., 2025). Concurrent work (Liang et al., 2025b; Xu et al., 2025) tackles real scenes but can not handle diverse generated scenes or large viewpoint changes.

4D generation. Recent years have seen rapid progress in 4D generation, i.e., dynamic 3D scene synthesis. Many approaches leverage input text prompts or images as guidance. Following the introduction of large-scale generative models for this task (Singer et al., 2023), subsequent works have made notable advances in both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a).

While text conditioning is widely used, alternative methods aim to lift 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Other efforts explore incorporating physics priors into generation pipelines (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025), or enhancing motion controllability with template-driven approaches (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b). A parallel direction emphasizes compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024).

Another strand of research extends 3D GAN frameworks into the 4D setting by training directly on 2D video data (Bahmani et al., 2023a; Xu et al., 2023). However, such methods are typically limited by small-scale, single-category datasets, which constrain generalization. Moreover, most existing approaches remain object-centric, often neglecting background elements. As a result, their overall visual fidelity lags behind the high photorealism achieved by recent video generation models, which our method leverages.

Camera-conditioned video models. Recent progress has focused on incorporating camera control into video diffusion models. The pioneering work MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Subsequent efforts (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) enhanced conditioning by representing cameras using Plücker coordinates, while others use follow MotionCtrl to simply use poses flattened (Wang et al., 2025b). Another line of research (Hu et al., 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) enabled camera motion control without introducing additional trainable parameters. Despite these advances, all of the above rely on U-Net-based architectures. Other approaches use low-rank approaches for camera control (Zhang et al., 2025b).

Several recent works continue to extend U-Net-based approaches for camera control (Zhao et al., 2024; Xu et al., 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024), while others begin exploring diffusion transformers. Cheong et al. (Cheong et al., 2024) propose one such transformer-based framework, though its scene and visual quality remain limited. DimensionX (Sun et al., 2024d) introduces joint space-time control in diffusion transformers, but relies on pre-defined, non-continuous camera trajectories. Other works (Huang et al., 2024a; Mai et al., 2025; Jiang et al., 2025b) instead explore pose and geometry estimation with a video DiT. CAT4D (Wu et al., 2025b) proposes a multi-view video diffusion model derived from fine-tuning a multi-view backbone. Large-

scale synthetic data generation further boosts performance, as demonstrated by SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a), which achieve strong results on camera-controlled video generation using transformer-based architectures. Finally, 4DiM (Watson et al., 2024) trains a space-time diffusion model from scratch for novel view synthesis from a single input image. Another line of work edits camera movements of videos with underlying motion using camera retargeting (Seo et al., 2025; Ma et al., 2025). One popular direction has been to use a spatio-temporal cache, e.g., using point clouds or optical flow, to condition the camera-controlled video generation (Wu et al., 2025c; Ren et al., 2025; Wang et al., 2025d; Cao et al., 2025; YU et al., 2025; Jin et al., 2025; Xing et al., 2025; Yan et al., 2025; Li et al., 2025b; Gu et al., 2025).